

Practical 10a.

```
niyathi@LAPTOP-NIYATHI:~/OSLab$ gcc practical10a.c -o practical10a
niyathi@LAPTOP-NIYATHI:~/OSLab$ ./practical10a
Multithreaded Counter Program
Number of threads: 4
Increments per thread: 100000

Expected counter value: 400000
Actual counter value: 197334

Race condition detected!
```

Practical 10b.

```
niyathi@LAPTOP-NIYATHI:~/OSLab$ gcc practical10b.c -o practical10b
niyathi@LAPTOP-NIYATHI:~/OSLab$ ./practical10b
=====
Multithreaded Counter - Mutex
=====
Number of threads      : 4
Increments per thread  : 100000

Expected counter value : 400000
Actual counter value   : 400000
Execution time         : 0.024804 seconds

Result: Correct!
Race condition prevented using mutex.
```